

Contents: *Stability Analysis in S domain*

- *Root Locus*

By *CHANDRA*



Control Systems

For stability of system k should be > 0

$$\therefore k > 0$$

$$\frac{0.65 - 0.115}{0.65} > 0$$

$$\frac{0.65}{0.1} > 15$$

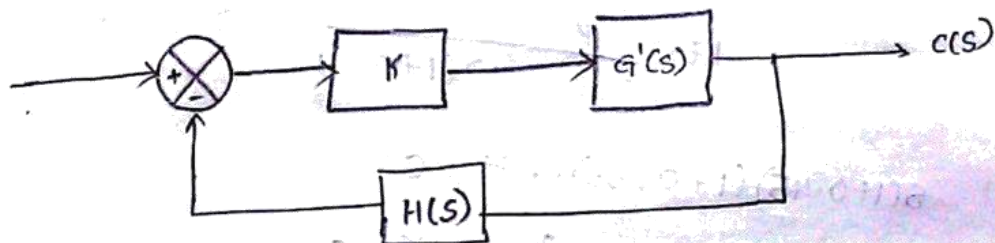
$$0 < k < 6.5$$

* Root Locus :

Movement of closed loop poles with small changes in the parameters of the system can be known by Root Locus method. This method is introduced by W.R. Evans. in 1948.

This is a graphical method in which movement of poles in the s -plane is sketched when a parameter of system is varied from 0 to ∞

* Concept of Root Locus :



Characteristic Eqn

$$1 + G(s)H(s) = 0.$$

$$\text{Where } G(s) = KG'(s)$$

$$\Rightarrow 1 + KG'(s)H(s) = 0.$$

The roots of the above equation are now dependent upon k . If k is varied from $-\infty$ to $+\infty$

then for each value of K , we will get separate set of locations of the roots of characteristic equation. If all such locations are joined, the resulting locus is called Root locus.

When K is varied from 0 to ∞ , the plot is called 'Direct Root locus' and when K is varied from $-\infty$ to ∞ , the plot obtained is 'Inverse Root locus'.

Ex :

consider unity feedback system with $G(s) = \frac{K}{s}$. Obtain the Root locus.

We have $1 + G(s)H(s) = 0$

$$\Rightarrow 1 + \frac{K}{s} = 0$$

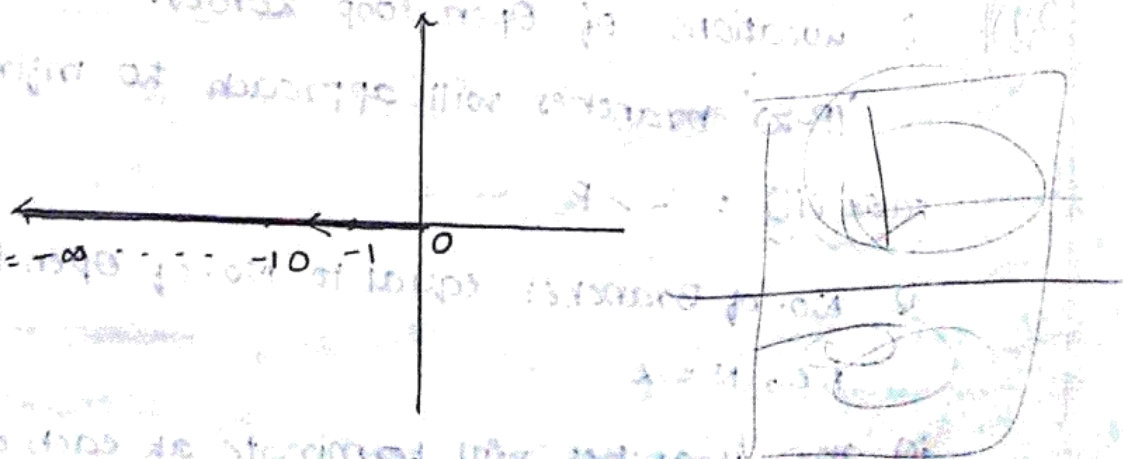
$$\Rightarrow K + s = 0$$

$$\Rightarrow s = -K$$

For $K=0$, $s=0$, for $K=100$, $s=-100$

$K=1$, $s=-1$

$K=10$, $s=-10$, $K=\infty$, $s=-\infty$.



* Rules of constructing a Root Locus :

1. The Root locus is always symmetrical about real axis.
2. The Roots of characteristic Equation are either Real or complex conjugate or combination of both. Therefore their locus must be symmetrical about Real axis of the s-plane.

3. Let $G(s)H(s)$ = open loop transfer function of the system where 'P' be no. of open loop poles and Z = no. of open loop zeroes.

case (i) : $P > Z$

i) the no. of branches 'N' equal to no. of open loop poles i.e., $N = P$.

ii) Branches will start from each of the location of open loop pole. out of 'P' no. of branches 'Z' no. of branches will terminate at the locations of open loop zeroes. Remaining ' $(P-Z)$ ' branches will approach to infinity (∞)

case (ii) : $Z > P$

i) No. of Branches equal to no. of Open loop zeroes i.e., $N = Z$.

ii) The Branches will terminate at each of the finite location of open loop zero. But out of 'Z' no. of branches, 'P' no. of branches will

start from each of the finite open loop pole, while remaining ' $(z-p)$ ' no. of branches will originate from infinity (∞) and will approach to finite zeroes.

case (iii): $P = Z$

i) The no. of branches $N = P = Z$, separate Branch will start from each of the open loop pole and terminate at open loop zero. No Branch will start or terminate at ∞ when $P = Z$.

3. A point on the Real axis lies on the root locus if the sum of number of open loop poles and zeroes on the real axis, to the right hand side of this point is odd.

4. The $(n-m)$ root locus branches that tend to ∞ , along the straight lines are called 'Asymptotes'. Angle of such Asymptotes are given by

$$\phi_A = \pm \frac{180^\circ [2q+1]}{(n-m)}$$

where $q = 0, 1, 2, \dots, (n-m)-1$

m = no. of zeroes,

n = no. of poles.

Asymptotes are always symmetrical about Real axis.

5. The point of Intersection of Asymptotes with the real axis is at $s = \sigma_A$, where

$$\sigma_A = \frac{(\text{sum of poles} - \text{sum of zeroes})}{(n-m)} = \text{centroid.}$$

6. The point from where branches break into complex from real is called Break-away point.

The point from where branches break into Real from complex is called Break-in point.

The Break-away and Break-in points of the Root Locus are determined from Roots of the Equation $dk/ds = 0$

If 'n' no. of branches of Root Locus meet at a point, then they break-away at an angle of $\pm \frac{180^\circ}{n}$

7. The angle of departure from a complex open loop pole is given by $\phi_p = \pm 180^\circ(2q+1) + \phi$

where $q = 0, 1, 2, \dots$

ϕ = net angle contribution at the pole by all other open loop poles and zeroes.

The angle of arrival at a complex open loop zero is given by $\phi_z = \pm 180^\circ(2q+1) + \phi$

where ϕ = net angle contribution at the zero by all other open loop poles and zeroes.

8. The Intersection of Root Locus branches with an Imaginary axis can be determined by use of Routh criterion (or) by letting $s = j\omega$ in the characteristic equation and equating Real part and Imaginary part to zero, to solve for ω & K . The value of ω is the intersection

point on imaginary axis and K is the value of gain at the intersection point.

a. Open Loop Gain $K = \frac{\text{Product of vector lengths from open loop poles to the point } s_a}{\text{Product of vector lengths from open loop zeroes to the point } s_a}$

1. A unity feedback control system has an open loop transfer function $G(s)H(s) = \frac{K}{s(s^2 + 4s + 13)}$. Sketch the root locus.

Sol:- Step-1 :

Locate the poles and zeroes of $G(s) \cdot H(s)$ on the s -plane. The root locus branch starts from open loop poles & terminates at zeroes.

To locate poles and zeroes,

$$s(s^2 + 4s + 13) = 0$$

$$s = 0, \quad s^2 + 4s + 13 = 0$$

$$s = \frac{-4 \pm \sqrt{16 - 4(13)}}{2}$$

$$s = -2 \pm \sqrt{4 - 13}$$

$$s = 0, \quad s = -2 \pm j3$$

The number of root locus branches is equal to no. of poles of open loop transfer function.

Step-2 :

Determine the Root locus on the real axis.

Step-3 :

To find angle of asymptotes and centroid and meeting point of asymptotes with real axis.

No. of Asymptotes is equal to no. of Root locus

branches going to ∞ .

$$\text{Angle of Asymptotes} = \frac{\pm 180^\circ(2q+1)}{n-m}$$

$$q = 0, 1, 2, \dots, [(n-m)-1]$$

$$n = \text{no. of poles} = 3$$

$$m = \text{no. of zeroes} = 0.$$

$$\Rightarrow q = 0, 1, 2$$

$$\Rightarrow \text{Angles} = \pm \frac{180^\circ}{3} = \pm 60^\circ, (q=0)$$

$$= \pm \frac{180^\circ(3)}{3} = \pm 180^\circ, (q=1)$$

$$= \pm \frac{180^\circ(5)}{3} = \pm 60^\circ(5) = \pm 300^\circ, (q=2)$$
$$= \pm 60^\circ$$

$$\text{centroid} = \frac{\text{sum of poles} - \text{sum of zeroes}}{n-m}$$

$$= \frac{0 - 2 + j3 - 2 - j3}{3 - 0} = \frac{-4}{3} = -1.33$$

Step-4 :

Find the break away & break in points.

These points either lie on real axis or exist as complex conjugate pairs. If there is a root locus on real axis b/w 2 poles, then there exists a

Break-away point.

Similarly, if there is a root locus on real axis b/w 2 zeroes, then there exists a Break-in point

and if there is a root locus on real axis b/w

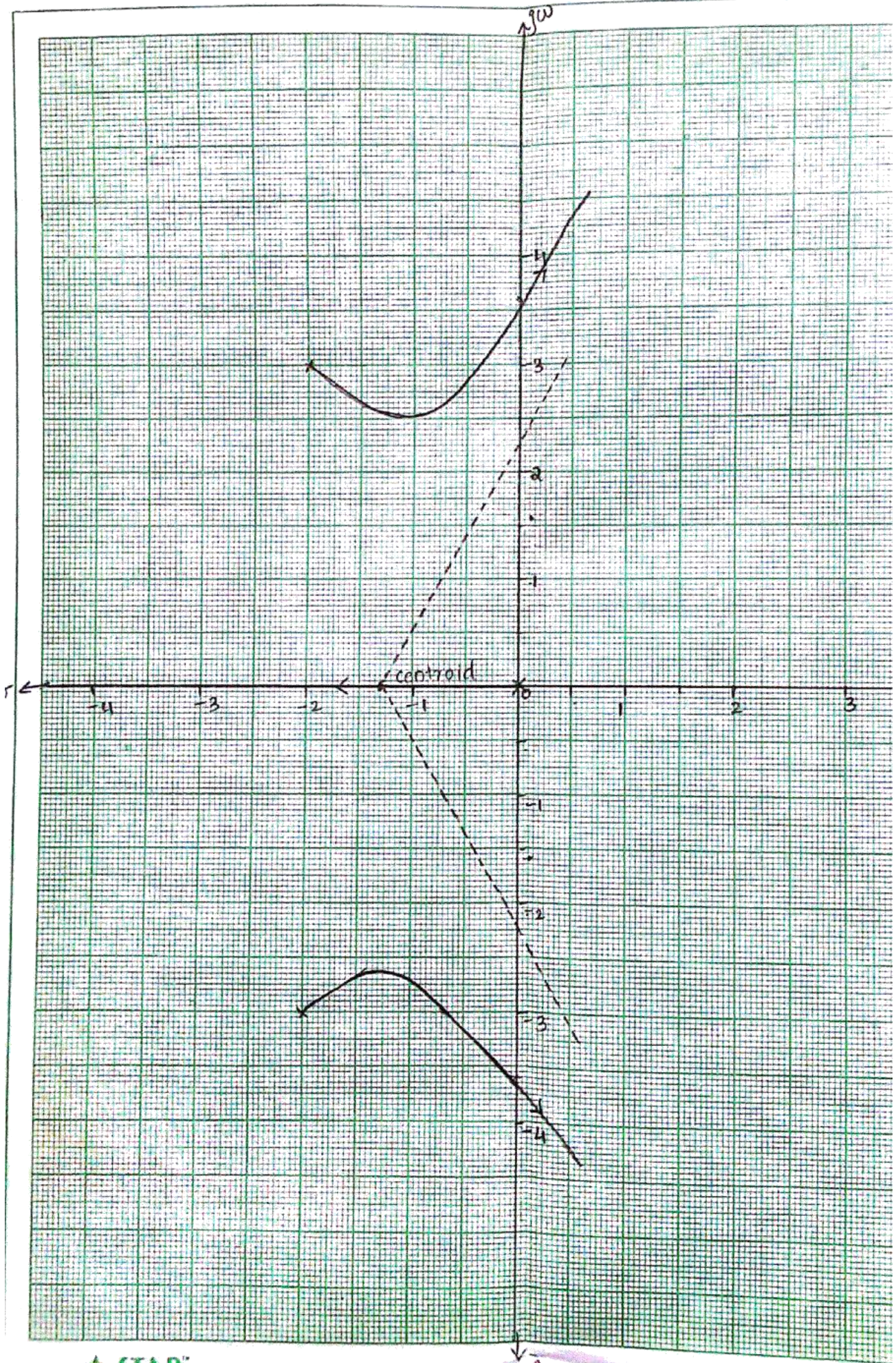
pole and zero, then there may be or may not be

Break away & Break in points.

branches going to ∞ .

Example-1 :

$$G(s)H(s) = \frac{s}{s(s^2 + 4s + 13)}$$



If $B(s) + K A(s) = 0$ is characteristic Equation.

$K = -\frac{B(s)}{A(s)}$ then K should be real & positive

then only we can say Breakaway & Breakin point exists.

$$\frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{K}{s(s^2 + 4s + 13)}}{1 + \frac{K}{s(s^2 + 4s + 13)}}$$

$$\frac{G(s)}{1 + G(s)H(s)} = \frac{K}{s(s^2 + 4s + 13) + K}$$

characteristic Eqn $s(s^2 + 4s + 13) + K = 0$

$$\Rightarrow K = -s(s^2 + 4s + 13)$$

$$\Rightarrow K = -s^3 - 4s^2 + 13s$$

$$\Rightarrow \frac{dK}{ds} = -3s^2 - 8s + 13$$

$$\frac{dK}{ds} = -(3s^2 + 8s + 13) = 0$$

$$\Rightarrow 3s^2 + 8s + 13 = 0$$

$$s = \frac{-8 \pm \sqrt{64 - 4(13)(3)}}{2(3)}$$

$$s = \frac{-4 \pm \sqrt{16 - 39}}{3}$$

$$s = \frac{-4}{3} \pm \frac{j\sqrt{23}}{3}$$

$$s = -1.33 \pm j1.6$$

$$\Rightarrow K = -s^3 - 4s^2 - 13s$$

$$K = -(-1.33 + j1.6)^3 - 4(-1.33 + j1.6)^2 - 13(-1.33 + j1.6)$$

K

$\frac{3.7}{2.3}$

K is neither positive nor real, hence Breakin
in Breakaway points does not exist

Step-5 :

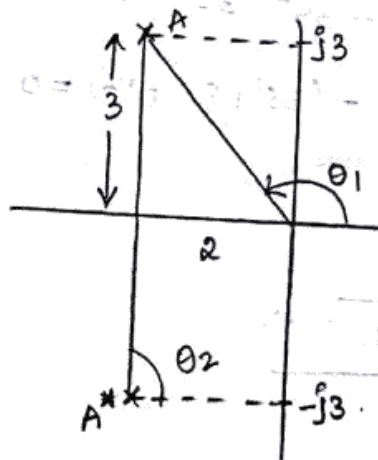
If there is a complex pole, then determine the angle of departure from the complex pole

If there is a complex zero, then determine the angle of arrival from the complex zero.

Angle of departure =

$180^\circ - (\text{sum of angles of vector to the complex pole } A \text{ from other poles}) + (\text{sum of angles of vector to the complex poles } A \text{ to zeroes})$

$$\phi_d = 180^\circ - \phi, \quad \phi = \sum \phi_p - \sum \phi_z$$



$$\theta_1 = 180^\circ - \tan^{-1}\left(\frac{3}{2}\right) = 123.7^\circ$$

$$\theta_2 = 90^\circ$$

$$\phi_d = 180^\circ - (\theta_1 + \theta_2)$$

$$(\text{Angle} = 180^\circ - (123.7^\circ + 90^\circ))$$

$$\text{Angle of departure} = 180^\circ - 213.7^\circ$$

$$\text{at } A) = -33.7^\circ$$

Angle of departure

$$\text{At } A^* = 33.7^\circ$$

Step-6 :

Find the points where the Root Loci may cross the imaginary axis by RH criterion.

$$\text{Characteristic eqn } s^3 + 4s^2 + 13s + K = 0$$

$$\begin{array}{c|cc} s^3 & 1 & 13 \\ s^2 & 4 & K \\ s & \frac{52-K}{4} & 0 \\ s^0 & K & \end{array}$$

$$\frac{52-K}{4} > 0$$

$$K = 52$$

$$\text{Auxiliary Eqn } 4s^2 + K = 0$$

$$4s^2 + 52 = 0$$

$$s^2 = -\frac{52}{4}$$

$$s = \pm j\sqrt{13}$$

$$s = \pm j3.6$$

$$2. \quad G(s) = \frac{K(s+9)}{s(s^2+4s+11)}$$

Sol: step-1 :

locate the poles and zeroes of $G(s)$ on the s -plane. The root locus branch starts from open loop poles and terminates at zeroes.

$$s(s^2+4s+11) = 0$$

$$s = 0, \quad s = \frac{-4 \pm \sqrt{16-4(11)}}{2}$$

$$s = -2 \pm \sqrt{4-11}$$

$$s = -2 \pm j\sqrt{7}$$

$$\text{poles : } s = 0, s = -2 \pm j2.6$$

$$\text{zeroes : } s+9=0, s=-9$$

The no. of root locus branches is equal to no of poles of open loop poles.

step-2 :

Determine the root locus on the real axis.

step-3 :

$$\text{Angle of Asymptotes} = \pm \frac{180^\circ(2q+1)}{n-m}$$

$$q = 0, 1, 2, \dots, [(n-m)-1]$$

$$q = 0, 1$$

$$\therefore n = 3$$

$$m = 1$$

$$\text{Angles} = \pm 90^\circ \text{ or } \pm 180^\circ \cdot 2 \neq 0$$

$$\text{centroid} = \frac{\text{sum of poles} - \text{sum of zeros}}{n-m}$$

$$= \frac{0 - 2 + j2 \cdot 6 - 2 - j2 \cdot 6 + 9}{2}$$

$$= \frac{-4 + 9}{2} = \frac{5}{2} = 2.5$$

step-4 :

$$\begin{aligned} \frac{G(s)}{1 + G(s)H(s)} &= \frac{\frac{K(s+9)}{s(s^2+4s+11)}}{1 + \frac{K(s+9)}{s(s^2+4s+11)}} \\ &= \frac{K(s+9)}{s(s^2+4s+11) + K(s+9)} \end{aligned}$$

Characteristic Eqn :

$$s(s^2+4s+11) + K(s+9) = 0$$

$$K = \frac{-s(s^2+4s+11)}{(s+9)}$$

$$\frac{dK}{ds} = K = \frac{-s^3 + 4s^2 - 11s}{s+9}$$

$$\frac{dK}{ds} = \frac{(s+9)[-3s^2 - 8s - 11] - [-s^3 + 4s^2 - 11s](1)}{(s+9)^2} = 0$$

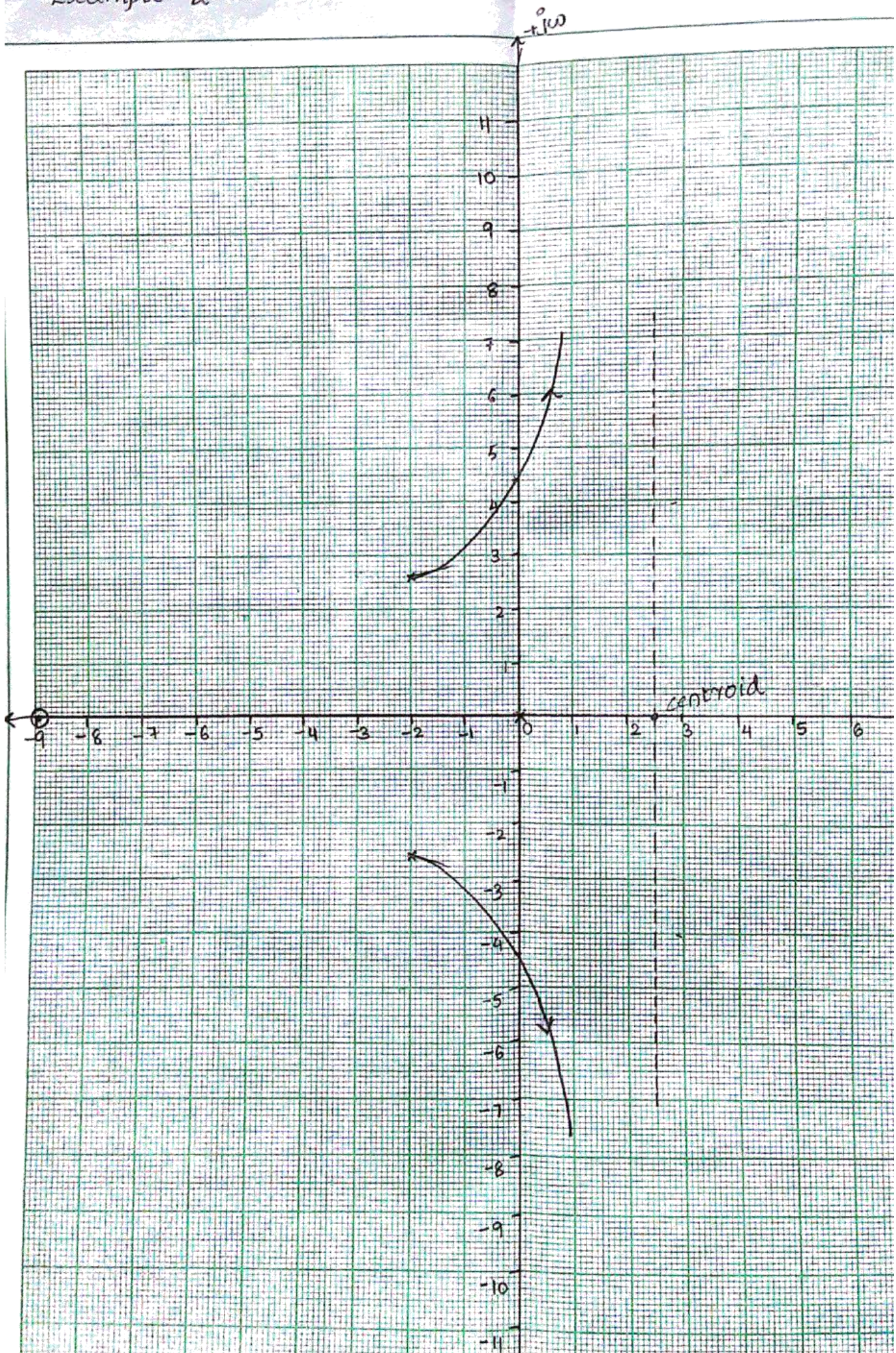
$$\Rightarrow -(3s^2 + 8s + 11)(s+9) + s^3 + 4s^2 + 11s = 0$$

$$\begin{aligned} &= s^3 - 3s^3 - 27s^2 - 8s^2 - 72s + 11s - 99 + \\ &\quad s^3 + 4s^2 + 11s = 0 \end{aligned}$$

$$\Rightarrow -2s^3 - 31s^2 - 72s - 99 = 0$$

Angles = $\pm 90^\circ$ or $\pm 180^\circ$. $z \neq 0$

Example - 2 :



$$\Rightarrow 2s^3 + 31s^2 + 72s + 99 = 0 \quad -2 + 31 - 72 + 99 = 0$$

$$s = -13.02, s = -1.2 + 1.5j, s = -1.2 - 1.5j$$

$$K = \frac{-s(s^2 + 4s + 11)}{s + 9}$$

$$= \frac{-13.02 [(-13.02)^2 + 4(-13.02) + 11]}{-13.02 + 9}$$

$$= -753$$

$\therefore K$ is not positive, Break-away & Break-in pts does not exist.

Step-5:

Angle of Departure

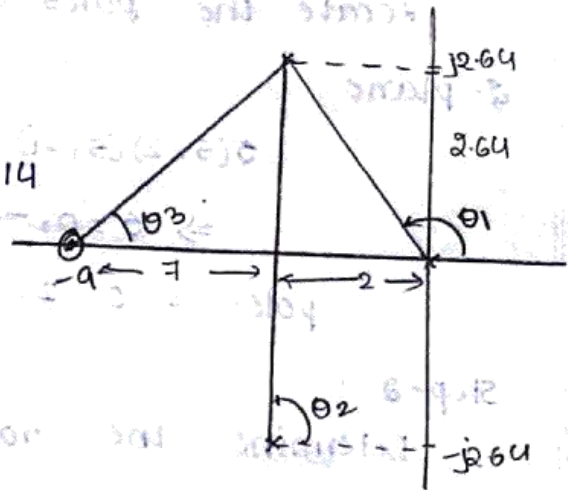
$$\phi_d = 180^\circ - (\theta_1 + \theta_2) + \theta_3$$

$$\theta_1 = 180^\circ - \tan^{-1}\left(\frac{2.64}{2}\right) = 127.14^\circ$$

$$\theta_2 = 90^\circ$$

$$\theta_3 = 180^\circ - \tan^{-1}\left(\frac{2.64}{7}\right)$$

$$= 159.33 \text{ to } 20.66^\circ$$



$$\phi_d = 180^\circ - (127.14 + 90^\circ) + 159.33 \text{ to } 20.66^\circ$$

$$\phi_d = -10.48^\circ \text{ at } A, A^* = 16.48^\circ$$

Step-6:

$$s^3 + 4s^2 + 11s + Ks + 9K = 0$$

$$s^3 + 4s^2 + s(11+K) + 9K = 0$$

$$s^3 \quad | \quad 1 \quad 11+K$$

$$s^2 \quad | \quad 4 \quad 9K$$

$$s^1 \quad | \quad \frac{4(11+K) - 9K}{4} \quad 0$$

$$s^0$$

$$\frac{4(11+K) - 9K}{4} > 0$$

$$44 + 4K - 9K > 0$$

$$44 - 5K > 0$$

$$5K < 44$$

$$K < 8.8$$

$$4s^2 + 9k = 0$$

$$4s^2 + 9(8) = 0$$

$$4s^2 = -72$$

$$s^2 = -18$$

$$s = \pm j4.24$$

3. sketch the root locus of the system whose open loop transfer function is $G(s) = \frac{K}{s(s+2)(s+4)}$. Find the value of K so that the damping ratio of closed loop system is 0.5. System is unity feedback system.

Sol:-

Step-1:

Locate the poles and zeroes of $G(s)$ on the s -plane

$$s(s+2)(s+4) = 0$$

$$\Rightarrow s = 0, -2, -4$$

$$\text{poles} = 0, -2, -4$$

Step-2:

Determine the root locus on the real axis.

Step-3:

$$\text{Angle of asymptotes} = \pm \frac{180^\circ (n-m)}{n-m}$$

$$n=3, m=0$$

$$q = 0, 1, 2 \dots [(n-m)-1]$$

$$q = 0, 1, 2$$

$$\text{for } q=0, \text{ angle} = \pm \frac{180^\circ (1)}{3} = \pm 60^\circ$$

$$q=1, \text{ angle} = \pm \frac{180^\circ (3)}{3} = \pm 180^\circ$$

$$q=2, \text{ angle} = \pm \frac{180^\circ (5)}{3} = \pm 300^\circ = \pm 60^\circ$$

$$\text{Centroid} = \frac{\text{sum of poles} - \text{sum of zeroes}}{(n-m)}$$

$$= \frac{0-2-4-0}{3}$$

$$= -\frac{6}{3} = -2$$

step-4 :

$$\frac{G(s)}{1+G(s)H(s)} = \frac{K}{s(s+2)(s+4)}$$

$$= \frac{K}{s(s+2)(s+4) + K}$$

charac Eqn

$$s(s+2)(s+4) + K = 0$$

$$\Rightarrow K = -(s^2+2s)(s+4)$$

$$\Rightarrow K = -(s^3+4s^2+2s^2+8s)$$

$$\Rightarrow K = -(s^3+6s^2+8s)$$

$$\frac{dK}{ds} = -3s^2 - 12s - 8$$

$$\frac{dK}{ds} = 0$$

$$\Rightarrow 3s^2 + 12s + 8 = 0$$

$$s = \frac{-12 \pm \sqrt{144 - 4(24)}}{2(3)}$$

$$s = \frac{-8 \pm 2\sqrt{10}}{2(3)}$$

$$s = \frac{-8 \pm 2\sqrt{10}}{3}$$

$$s = -0.55, -4.77$$

$$s = -0.84, -3.154$$

$$= -(s^3+6s^2+8s)$$

$$= -[(-0.84)^3 + 6(-0.84)^2 + 8(-0.84)]$$

$$= 3.07$$

$$\text{for } s = -3.154$$

$$K = -3.07$$

for $s = -0.84$, K is real and positive.

$\therefore$ there are no complex roots, angle of departure is not required.

Step-5:

$$s^3 + 6s^2 + 8s + K = 0$$

$$\begin{array}{r|rr} s^3 & 1 & 8 \\ s^2 & 6 & K \\ s^1 & \frac{48-K}{6} & 0 \\ s^0 & K & \end{array}$$

$$\frac{48-K}{6} > 0$$

$$48 - K > 0$$

$$K < 48$$

Auxiliary Eqn $6s^2 + K = 0$.

$$6s^2 + 48 = 0$$

$$s^2 = -\frac{48}{6}$$

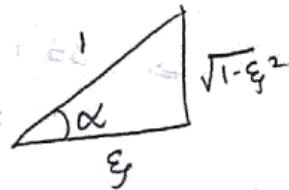
$$s^2 = -8$$

$$s = \pm j2.828$$

To find the value of K corresponding to $\xi = 0.5$.

$$\cos \alpha = \frac{\xi}{1}$$

$$\begin{aligned} \Rightarrow \alpha &= \cos^{-1}(\xi) \\ &= \cos^{-1}(0.5) \\ &= 60^\circ \end{aligned}$$



Open loop gain $K = \frac{\text{Product of vector lengths from open loop poles to point } s_a}{\text{Product of vector lengths from open loop zeroes to point } s_a}$

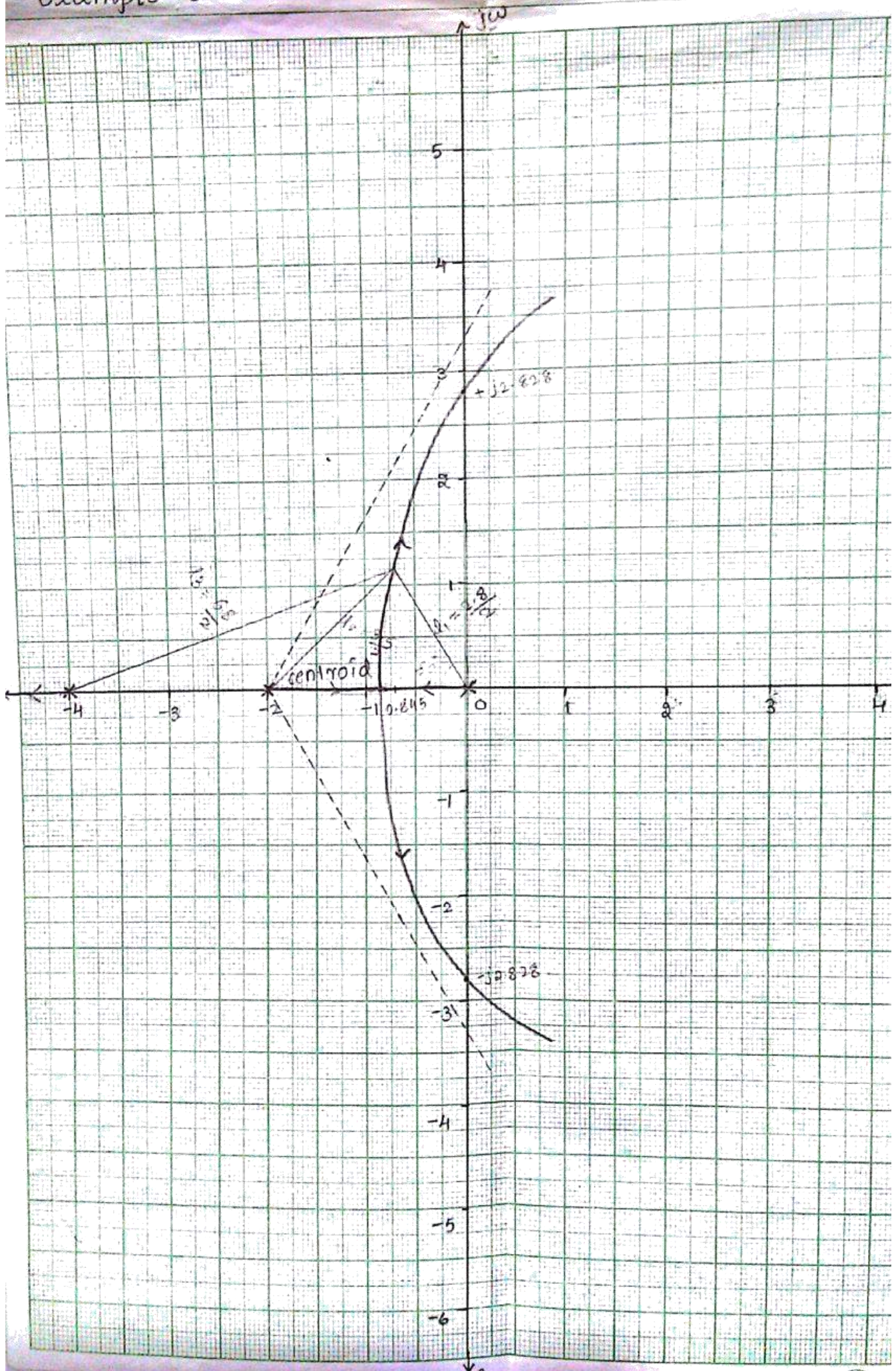
Product of vector lengths from open loop zeroes to point s_a .

$$d_1 = \frac{2.8}{2} = 1.4, \quad d_2 = \frac{3.5}{2} = 1.75, \quad d_3 = \frac{6.8}{2} = 3.4$$

$$K = \frac{d_1 d_2 d_3}{1} = (1.4)(1.75)(3.4) = 8.33 \approx 8$$

Example - 3 :

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Complete root locus has 3 branches.
 one branch starts at a pole at $s = -4$ and travel through -ve real axis to meet the zero at ∞ .
 The other two root locus branches starts at $s = 0$ & $s = -2$ and travel through -ve real axis, break away from real axis at $s = -0.845$, then crosses imaginary axis at $s = \pm j2.8$ and travel parallel to asymptotes to meet the zeroes at ∞ .

4. The open loop transfer function of unity feedback control system is given by $G(s) = \frac{k(s+1)}{s^2}$, sketch the root locus.

Sol:

Step-1 :

poles : $s^2 = 0$
 $\Rightarrow s = 0, 0$

zeros : $s+1 = 0$
 $s = -1$

locate poles and zeroes on s-plane

Step-2 :

Determine the root locus on the real axis.

Step-3 :

Angle of Asymptotes = $\pm \frac{180^\circ(2q+1)}{n-m}$

$n = 2, m = 1$

$\Rightarrow q = 0, 1, 2, \dots, [(n-m)-1]$

$= 0, 1, 2, \dots, [(2-1)-1]$

$q = 0$

Angle for $q=0 = \pm \frac{180^\circ(1)}{2-1}$
 $= \pm 180^\circ$

Centroid = $\frac{\text{sum of poles} - \text{sum of zeroes}}{n-m}$

$= \frac{0 + 0 + 1}{2-1} = -0.5$

step-4

$$\frac{G(s)}{1+G(s)H(s)} = \frac{\frac{k(s+1)}{s^2}}{1 + \frac{k(s+1)}{s^2}} = \frac{k(s+1)}{s^2 + k(s+1)}$$

$$s^2 + k(s+1) = 0.$$

$$s^2 + ks + k = 0.$$

$$k = \frac{-s^2}{s+1}.$$

$$\frac{dk}{ds} = 0$$

$$\Rightarrow \frac{(s+1)(-2s) - (-s^2)(1)}{(s+1)^2} = 0$$

$$\Rightarrow -2s(s+1) + s^2 = 0$$

$$\Rightarrow -2s^2 - 2s + s^2 = 0.$$

$$\Rightarrow -s^2 - 2s = 0$$

$$\Rightarrow s^2 + 2s = 0$$

$$\Rightarrow s = 0, -2.$$

for $s = 0$, $k = 0$

for $s = -2$, $k = \frac{-4}{-2+1} = 4$

for $s = -2$, k is real & positive.

$\therefore$ there are no complex roots, angle of departure does not exist.

$$s^2 + ks + k = 0$$

s^2	1	k
s^1	k	0
s^0	$\frac{k^2}{k} = k$	

$$k > 0.$$

$$ks = 0$$

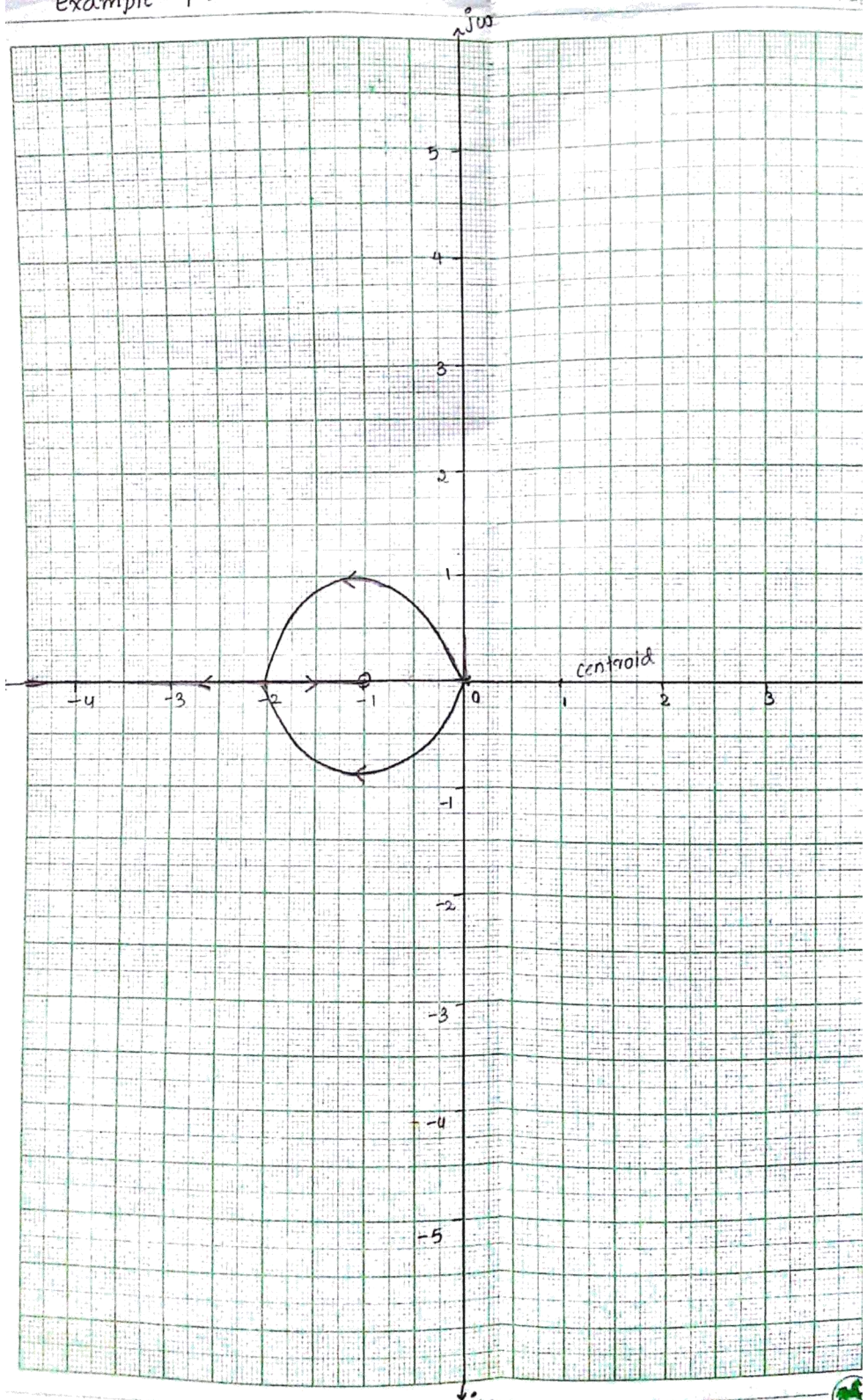
$$s = 0.$$

5.

$$G(s)H(s) = \frac{k(s+1)}{s(s-1)(s^2+4s+16)}$$

Example - 4 :

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* Effect of addition of open loop poles and zeroes:

Addition of poles:

1. Root locus shifts towards imaginary axis.
2. System stability relatively decreases.
3. System becomes more oscillatory in nature.
4. Range of operating values of K for stability of the system decreases.

Addition of zeroes:

1. Root locus shifts to left away from imaginary axis.
2. Relative stability of the system increases.
3. System becomes less oscillatory.
4. Range of operating values of K for system stability increases.

Ex 30

$$G(s)H(s) = \frac{K(s+1)}{s(s-1)(s^2+4s+16)}$$

poles : $s(s-1)(s^2+4s+16) = 0$

$$s=0, s=1, s^2+4s+16=0$$

$$s = -2 \pm 2\sqrt{3}j$$

$$s = -2 \pm 3.46j$$

zeroes : $s+1=0$
 $s=-1$

Step 1 : $G(s)H(s) = \frac{K(s+1)}{s(s-1)(s^2+4s+16)}$

Determine the root locus on the real axis

Step 2 :

$$\text{Angle of Asymptotes} = \pm \frac{180^\circ(2q+1)}{n-m}$$

$$n = 4, m = 1$$

$$q = 0, 1, 2 \quad (n-m)-1$$

$$(4-1)-1$$

$$4-2$$

$$q = 0, 1, 2$$

$$\text{for } q=0, \text{ angle} = \pm \frac{180^\circ(1)}{4-1} = \pm 60^\circ,$$

$$\text{for } q=1, \pm \frac{180^\circ(3)}{3} = \pm 180^\circ$$

$$\text{for } q=2, \pm \frac{180^\circ(5)}{3} = \pm 300^\circ = \mp 60^\circ.$$

$$\text{centroid} = \frac{\text{sum of poles} - \text{sum of zeros}}{n-m}$$

$$= \frac{0+1-2-j3 \cdot 4 - 2 + j3 \cdot 6 + 1}{3}$$

$$= \frac{-2}{3} = -0.66.$$

step-4 :

$$\frac{q(s)}{1+q(s)h(s)} = \frac{\frac{k(s+1)}{s(s-1)(s^2+4s+16)}}{1 + \frac{k(s+1)}{s(s-1)(s^2+4s+16)}}$$

$$= \frac{k(s+1)}{s(s-1)(s^2+4s+16) + k(s+1)}.$$

$$s(s-1)(s^2+4s+16) + k(s+1) = 0$$

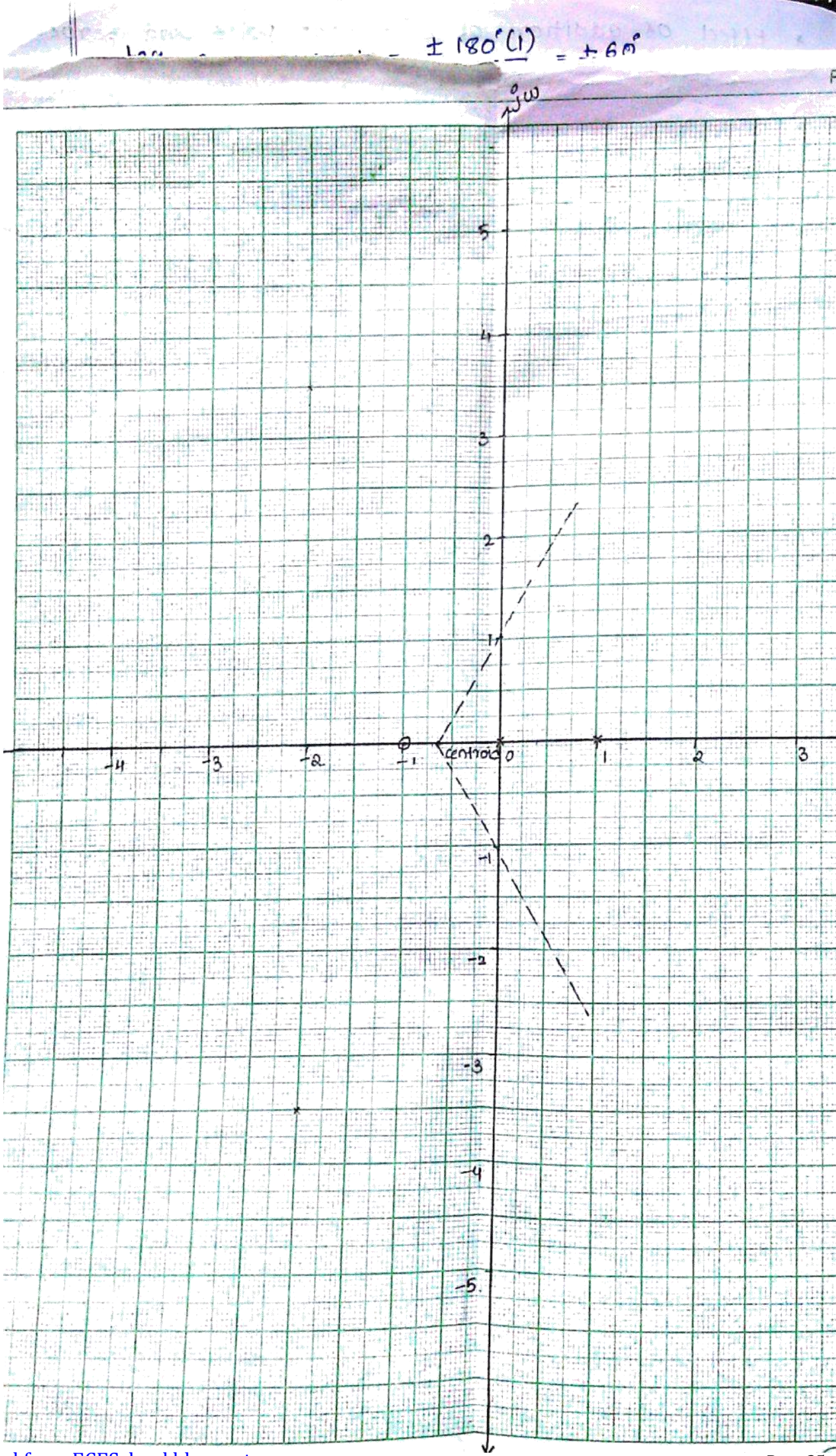
$$k = \frac{(s^2-s)(s^2+4s+16)}{(s+1)}$$

$$k = \frac{-(s^4 + 4s^3 + 16s^2 - s^3 - 4s^2 - 16s)}{s+1}$$

$$k = - \frac{(s^4 + 3s^3 + 12s^2 - 16s)}{s+1}.$$

$$\frac{dk}{ds} = 0 \Rightarrow \frac{(s+1)[-(4s^3 + 9s^2 + 24s - 16)] + (s^4 + 3s^3 + 12s^2 - 16s)}{(s+1)^2} = 0$$

$$\Rightarrow (s+1)(4s^3 + 9s^2 + 24s - 16) + s^4 + 3s^3 + 12s^2 + 16s = 0.$$



$$\Rightarrow 4s^4 + 6s^3 + 24s^2 - 16s + 4s^3 + 6s^2 + 24s - 16 +$$

$$s^4 + 3s^3 + 12s^2 - 16s = 0.$$

$$\rightarrow 5s^4 + 9s^3 + 36s^2 - 32s - 16 = 0.$$

$$\Rightarrow 4s^4 + 9s^3 + 24s^2 - 16s + 4s^3 + 9s^2 + 24s - 16 - s^4 - 3s^3 - 12s^2 + 16s = 0$$

$$\Rightarrow 3s^4 + 10s^3 + 21s^2 + 24s - 16 = 0$$

Contents: *Stability Analysis in S domain*

- *Routh's Stability Criterion*

By *CHANDRA*



Control Systems

UNIT- 3

STABILITY ANALYSIS IN S-DOMAIN

The term stability refers to stable working condition of a control system.

- Every working system is designed to be stable.
- In a stable system the response (or) output is predictable, finite and stable for a given input.
- A system is stable if its output is bounded (finite) for any bounded input.
- A system is asymptotically stable if in the absence of i/p, the o/p tends towards '0', irrespective of initial conditions.
- A system is stable, if for a bounded disturbing i/p signal, the o/p vanishes ultimately as $t \rightarrow \infty$.
- The system is unstable, if for a bounded disturbing i/p signal, the o/p is of infinite amplitude (or) oscillatory.
- For a bounded i/p signal if the o/p has constant amplitude oscillations then the system may be stable (or) unstable under some limited constraints.
- Such a system is called "limitedly stable".
- If a system o/p is stable, for all variations of its parameters then the system is called "absolutely stable system".
- If a system o/p is stable, for a limited range of variations of its parameters, then the system is called "conditionally stable system".

* Bounded input Bounded output stability [BIBO]:

$$\frac{C(s)}{R(s)} = M(s)$$

$$\Rightarrow M(s) = \frac{C(s)}{R(s)}$$

By applying I.L.T

$$\mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1}[M(s) R(s)] = m(t) * r(t)$$

$$c(t) = \int_{-\infty}^{\infty} m(\tau) r(t-\tau) d\tau$$

- A linear relaxed system is said to be having BIBO stability if every bounded i/p results in a bounded o/p.
- A Test for BIBO stability can be obtained from convolution Theorem,

$$c(t) = \int_0^{\infty} m(\tau) r(t-\tau) d\tau$$

- If $r(t)$ is bounded then there exist a constant A , such that
 $|r(t)| \leq A, \forall t$
- The condition for bounded o/p for this bounded i/p condition can be derived as follows.

$$|c(t)| = \left| \int_0^{\infty} m(\tau) r(t-\tau) d\tau \right|$$

$$\leq \int_0^{\infty} |m(\tau) r(t-\tau)| d\tau$$

$$\leq \int_0^{\infty} |m(\tau)| |r(t-\tau)| d\tau$$

$$\leq \int_0^{\infty} |m(\tau)| A d\tau$$

$$= A \int_0^{\infty} |m(\tau)| d\tau$$

If o/p $c(t)$ is bounded, then there exists a constant A_2 , such that

$$|c(t)| \leq A_2 \leq \infty$$

$$\therefore A_1 \int_0^{\infty} |m(\tau)| d\tau \leq A_2 < \infty$$

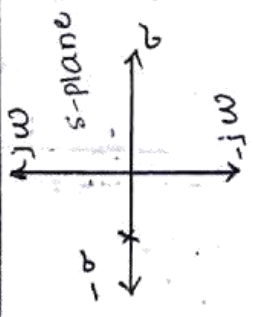
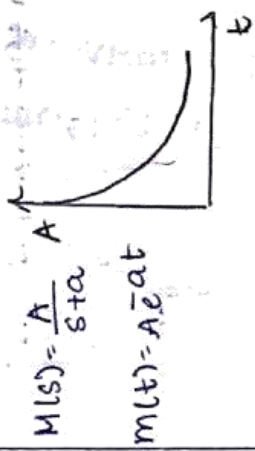
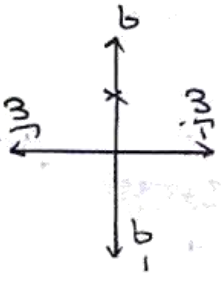
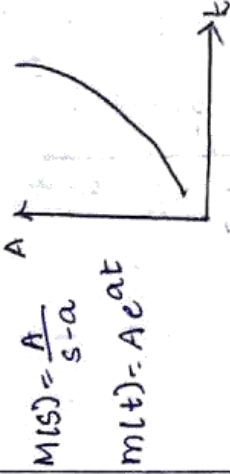
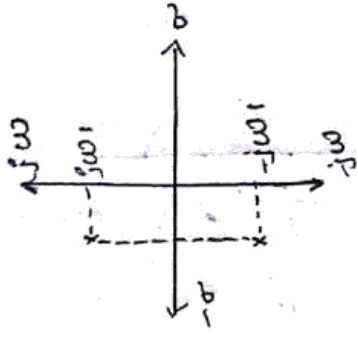
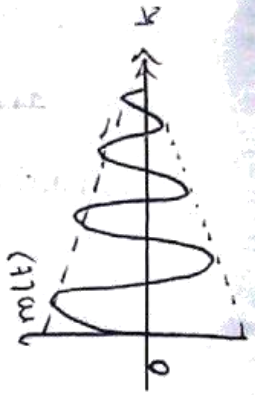
Above condition is satisfied if $\int_0^{\infty} |m(\tau)| d\tau$ is less than ∞ .

Hence for bounded o/p

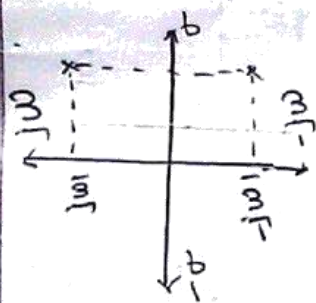
$$\int_0^{\infty} |m(\tau)| d\tau < \infty$$

$\therefore$ A system with impulse response $m(\tau)$ is BIBO stable, if and only if the impulse response is absolutely integrable i.e., $\int_0^{\infty} |m(\tau)| d\tau$ is finite. This means that, the area under the absolute value curve of $m(\tau)$, is evaluated from, $t=0$ to $t=\infty$ must be finite.

Location of roots on the s-plane for stability

S.No.	Nature of closed loop pole	Location	Response	Stability condition
1.	Root on -ve real axis		$M(s) = \frac{A}{s+a}$ $m(t) = A e^{-at}$ 	$\int_0^{\infty} m(\tau) d\tau$ is finite Hence system is stable,
2.	Root on +ve real axis		$M(s) = \frac{A}{s-a}$ $m(t) = A e^{at}$ 	$\int_0^{\infty} m(\tau) d\tau = \infty$ Hence system is unstable.
3.	Complex conjugate roots on left half of s-plane		$M(s) = \frac{A}{s - (-\sigma + j\omega)} + \frac{A}{s - (-\sigma - j\omega)}$ $m(t) = 2A e^{-\sigma t} \sin(\omega t + 90^\circ)$ 	$\int_0^{\infty} m(\tau) d\tau = \text{finite}$ Hence system is stable.

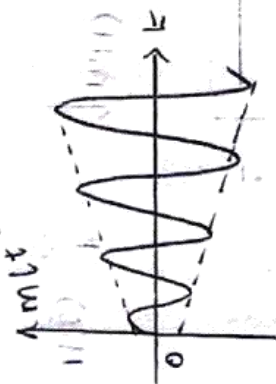
4. complex conjugate roots on right half of s-plane



$$M(s) = \frac{k}{s + (\sigma + j\omega)} + \frac{k}{s - (\sigma - j\omega)}$$

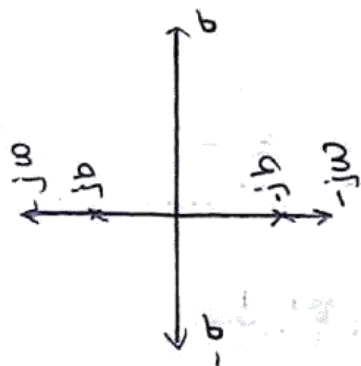
$$m(t) = 2ke^{\sigma t} \cos \omega t$$

$$= 2ke^{\sigma t} \sin(\omega t + 90^\circ)$$



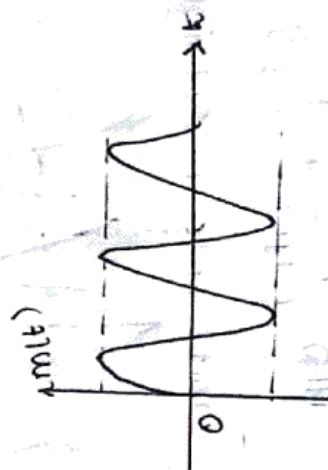
unstable.

5. single pair of roots on imaginary axis



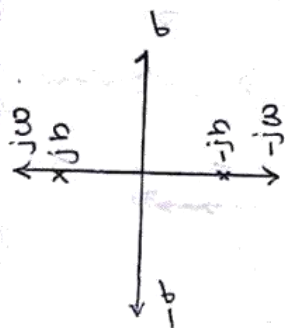
$$M(s) = \frac{k}{s + jb} + \frac{k}{s - jb}$$

$$m(t) = 2k \cos bt$$



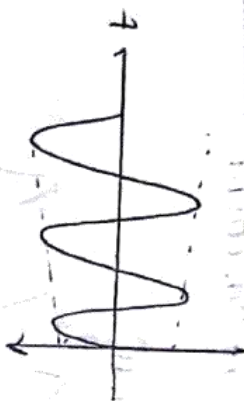
unstable.

6. Double pair of roots on imaginary axis



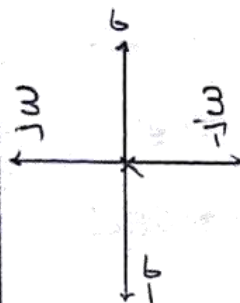
$$M(s) = \frac{k}{(s+jb)^2} + \frac{k}{(s-jb)^2}$$

$$m(t) = 2kt \cos bt = 2kt \sin(bt + 90^\circ)$$



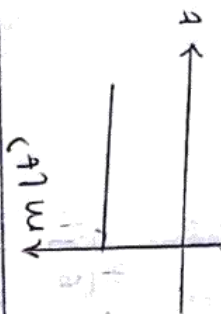
unstable.

7. single pole at origin



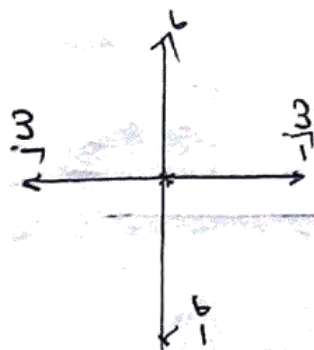
$$M(s) = \frac{k}{s}$$

$$m(t) = kt$$



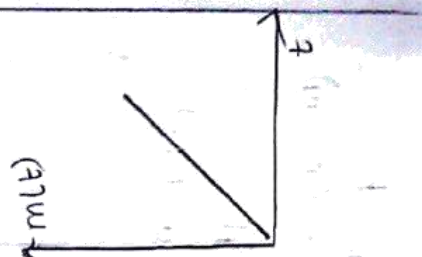
unstable.

8. Double pole at origin



$$M(s) = \frac{k}{s^2}$$

$$m(t) = kt^2$$



unstable.

* Hurwitz's Criterion :

The sufficient condition for having all the roots of characteristic equation in left half of s-plane is given by Hurwitz. It is referred to as Hurwitz's criterion.

It states that the necessary and sufficient condition to have all the roots of characteristic eqn in left half of s-plane is that the sub-determinants obtained from Hurwitz's determinant 'H' must be positive.

Method of forming Hurwitz's Determinant :

Characteristic Equation :

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

The Determinant

$$H = \begin{vmatrix} a_1 & a_3 & a_5 & \dots & a_{2n-1} \\ a_0 & a_2 & a_4 & \dots & a_{2n-2} \\ 0 & a_1 & a_3 & \dots & 0 \\ 0 & a_0 & a_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_n \end{vmatrix}$$

From Hurwitz's Determinants, the sub-determinants must be formed as

$$D_1 = |a_1|$$

$$D_2 = \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}$$

$$D_3 = \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix}, \dots, D_n = |H|$$

Adv & Disad of Hurwitz

1. Determine the stability of the given characteristic eqn by Hurwitz's method $H(s) = s^3 + s^2 + s + 4 = 0$.

Sol:-

Given $F(s) = s^3 + s^2 + s + 4$

$a_0 = 1, a_1 = 1, a_2 = 1, a_3 = 4$

order of the $F(s) = 3 = n$

$D_1 = |a_0| = 1$

$D_2 = \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} = \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} = 1 - 4 = -3$

$H = D_3 = \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 4 \end{vmatrix}$

$= 1[4] - 4[4]$

$= 4 - 16 = -12$

Since the determinants are not positive, so the system is unstable.

* ROUTH'S STABILITY CRITERION:

(or) Routh's Hurwitz's (or) RH Criterion.

In this criterion, the coefficients of

characteristic equation are arranged as rows in an array.

For example, consider a characteristic equation of

order- n $a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$

Routh array is formed as

s^n	a_0	a_2	a_4	a_6	...
s^{n-1}	a_1	a_3	a_5	a_7	...
s^{n-2}	b_1	b_2	b_3	...	
s^{n-3}	c_1	c_2	c_3	...	
$\vdots$	$\vdots$				
s^1	f_1				
s^0	g_1				

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}, \quad b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}, \quad c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

It states that for a system to be stable, it is necessary and sufficient that each term of first column of Routh array of its characteristic equation be positive. If this condition is not met, the system is unstable and the number of sign changes of the terms of the first column of the Routh array corresponds to the number of roots of characteristic eqn in the right half of s-plane.

1. Using Routh's criterion, determine the stability of the system represented by the characteristic equation $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$. Comment on the location of the roots of characteristic Equation.

Sol: Given $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

$$\Rightarrow a_0 = 1, a_1 = 8, a_2 = 18, a_3 = 16, a_4 = 5$$

s^4	1	18	5
s^3	8	16	0
s^2	16	5	
s^1	13.5	0	
s^0	5		

Since there all the elements in the first column are positive. hence the system is stable. All the roots lies in the left half of s-plane.

2. $s^6 + 2s^5 + 2s^4 + 3s^3 + 5s^2 + 6s + 1 = 0$

$a_0=1, a_1=2, a_2=2, a_3=3, a_4=5, a_5=6, a_6=1$

s^6	1	2	5	1
s^5	2	3	6	0
s^4	0.5	2	1.5	
s^3	-5	2	0	
s^2	+2.2	1	0	
s^1	4.2	0		
s^0	1			

Since some of the elements in first column are -ve, the system is unstable.
 $\therefore$ the sign changes 2 times, 2 roots lie in the right side of s-plane.

3. $s^3 + 6s^2 + 11s + 6 = 0$

$a_0=1, a_1=6, a_2=11, a_3=6$

s^3	1	11	
s^2	6	6	
s^1	10	0	
s^0	6		

Since all the elements of first column are positive, the system is stable.

$$4. s^3 + 4s^2 + s + 16 = 0.$$

$$a_0 = 1, a_1 = 4, a_2 = 1, a_3 = 16.$$

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 4 & 16 \\ s^1 & -3 & 0 \\ s^0 & +16 & \end{array}$$

All the elements present in the 1st column are not positive, system is unstable.

Since the sign changes two times, there are two roots present in the right side of the s-plane.

* Special cases of R-H criterion : Case-1

Ex: Method-1:

$$\text{Ex: } s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0.$$

$$a_0 = 1, a_1 = 2, a_2 = 3, a_3 = 6, a_4 = 2, a_5 = 1.$$

$$\begin{array}{c|ccc} s^5 & 1 & 3 & 2 \\ s^4 & 2 & 6 & 1 \\ s^3 & 0 & 1.5 & \\ s^2 & \infty & & \\ s^1 & \infty & & \\ s^0 & \infty & & \end{array}$$

$$\begin{array}{c|ccc} s^5 & 1 & 3 & 2 \\ s^4 & 2 & 6 & 1 \\ s^3 & \epsilon & 1.5 & 0 \\ s^2 & \frac{6\epsilon-3}{\epsilon} & 1 & 0 \\ s^1 & \frac{1.5(\frac{6\epsilon-3}{\epsilon})-\epsilon}{\frac{6\epsilon-3}{\epsilon}} & & \\ s^0 & 1 & & \end{array}$$

To examine sign changes

$$\lim_{\epsilon \rightarrow 0} \frac{6\epsilon-3}{\epsilon} = -\infty$$

$$\lim_{\epsilon \rightarrow 0} \frac{1.5\left(\frac{6\epsilon-3}{\epsilon}\right) - \epsilon}{\left(\frac{6\epsilon-3}{\epsilon}\right)} =$$

$$1.5 - \epsilon \left(\frac{\epsilon}{6\epsilon-3} \right) = 1.5.$$

$$\Rightarrow \begin{array}{c|ccc} s^5 & 1 & 3 & 2 \\ s^4 & 2 & 6 & 1 \\ s^3 & 0 & 1.5 & 0 \\ s^2 & -\infty & 1 & 0 \\ s^1 & 1.5 & 0 & 0 \\ s^0 & 1 & 0 & 0 \end{array}$$

$\therefore$ There are 2 sign changes, the system is unstable. $\emptyset$

$\Rightarrow$ 2 roots lie in right side of s-plane.

Method-2:

Ex: $s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0$

$$\begin{array}{c|ccc} s^5 & 1 & 3 & 2 \\ s^4 & 2 & 6 & 1 \\ s^3 & 0 & 1.5 & 0 \\ s^2 & \infty & & \\ s^1 & & & \\ s^0 & & & \end{array}$$

Replace s by $1/z$ in the eqn

$$\Rightarrow \frac{1}{z^5} + \frac{2}{z^4} + \frac{3}{z^3} + \frac{6}{z^2} + \frac{2}{z} + 1 = 0$$

$$\Rightarrow z^5 + 2z^4 + 6z^3 + 3z^2 + 2z + 1 = 0$$

$$\begin{array}{c|ccc} z^5 & 1 & 6 & 2 \\ z^4 & 2 & 3 & 1 \\ z^3 & 4.5 & 1.5 & 0 \\ z^2 & 2.33 & 1 & 0 \\ z^1 & -0.43 & 0 & 0 \\ z^0 & 1 & 0 & 0 \end{array}$$

Since, there are 2 sign changes, the system is unstable

$\rightarrow$ 2 roots lie in right side of s-plane.

case - 2 :

Ex: $s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$

Find the no. of roots of this eqn with +ve real part, zero real part & -ve real part

$a_0 = 1, a_1 = 4, a_2 = 3, a_3 = 0, a_4 = -16, a_5 = -64, a_6 = -48$

$$s^6 \mid \begin{array}{cccc} 1 & 3 & -16 & -48 \end{array}$$

$$s^5 \mid \begin{array}{cccc} 4 & 0 & -64 & 0 \end{array}$$

$$s^4 \mid \begin{array}{cccc} 3 & 0 & -48 & 0 \end{array} \Rightarrow A(s) = 3s^4 - 48 = 0$$

$$s^3 \mid \begin{array}{cccc} 0 & 0 & 0 & 0 \end{array} \quad \frac{dA}{ds} = 12s^3$$

$$s^2$$

$$s^1$$

$$s^0$$

$$\Rightarrow s^6 \mid \begin{array}{cccc} 1 & 3 & -16 & -48 \end{array}$$

$$s^5 \mid \begin{array}{cccc} 4 & 0 & -64 & 0 \end{array}$$

$$s^4 \mid \begin{array}{cccc} 3 & 0 & -48 & 0 \end{array}$$

$$s^3 \mid \begin{array}{cccc} 12 & 0 & 0 & 0 \end{array}$$

$$s^2 \mid \begin{array}{cccc} 0 & -48 & 0 & 0 \end{array}$$

$$s^1 \mid \begin{array}{cccc} +48 & 12 & 0 & 0 \end{array}$$

$$s^0 \mid \begin{array}{cccc} -48 & 0 & 0 & 0 \end{array}$$

$$\text{At } E \rightarrow 0 \quad \frac{+48(12)}{E} = +\infty$$

$$\Rightarrow s^6 \mid \begin{array}{cccc} 1 & 3 & -16 & -48 \end{array}$$

$$s^5 \mid \begin{array}{cccc} 4 & 0 & -64 & 0 \end{array}$$

$$s^4 \mid \begin{array}{cccc} 3 & 0 & -48 & 0 \end{array}$$

$$s^3 \mid \begin{array}{cccc} 12 & 0 & 0 & 0 \end{array}$$

$$s^2 \mid \begin{array}{cccc} 0 & -48 & 0 & 0 \end{array}$$

$$s^1 \mid \begin{array}{cccc} \infty & 0 & 0 & 0 \end{array}$$

$$s^0 \mid \begin{array}{cccc} -48 & 0 & 0 & 0 \end{array}$$

since the sign is changed by 1 time, the system is unstable & one root lies in the right hand side of s-plane

To find the dominant roots $A(s)=0$ is to be solved.

$$\Rightarrow A(s) = 3s^4 - 48 = 0$$

$$3s^4 = 48$$

$$s^4 = 16$$

$$s = \pm \sqrt{4}$$

$$\Rightarrow s = \pm 2, s = \pm 2j$$

$+2$ is in the right side of s -plane & the remaining roots lie in the left half of s -plane.

Ex. Find range of values of k , so that system with the following characteristic eqn will be stable.

$$F(s) = s(s^2 + s + 1)(s + 4) + k = 0$$

Sol:- $F(s) = s(s^2 + s + 1)(s + 4) + k = 0$

$$(s^3 + s^2 + s)(s + 4) + k = 0$$

$$s^4 + 4s^3 + s^3 + 4s^2 + s^2 + 4s + k = 0$$

$$\Rightarrow s^4 + 5s^3 + 5s^2 + 4s + k = 0$$

s^4	1	5	k
s^3	5	4	0
s^2	4.2	$\frac{5k}{5} = k$	0
s^1	$\frac{16.8 - 5k}{4.2}$		0
s^0	k		

For the system to be stable

$$\frac{16.8 - 5k}{4.2} > 0$$

$$16.8 - 5k > 4.2 \times 0$$

$$-5k > 4.2 - 16.8$$

$$-5k > -12.6 \quad | \times 16.8$$

$$k > 3.36$$

$\therefore$ The k should lies b/w

$$0 \text{ \& } 3.36 \Rightarrow 0 < k < 3.36$$

$$\text{A } k > 0$$

Ex: The equation is $s^4 + 3s^3 + 3s^2 + 2s + K = 0$. And the value of K for the system to be stable -

Sol: $F(s) = s^4 + 3s^3 + 3s^2 + 2s + K = 0$

$$\begin{array}{c|ccc} s^4 & 1 & 3 & K \\ \hline s^3 & 3 & 2 & 0 \\ \hline s^2 & 2.33 & K & \\ \hline s^1 & \frac{4.66-3K}{2.33} & 0 & \\ \hline s^0 & K & 0 & \end{array}$$

For the system to be stable,

$$\frac{4.66-3K}{2.33} > 0$$

$$4.66 - 3K > 0$$

$$4.66 < 3K$$

$$K > \frac{4.66}{3}$$

$$K > 1.553$$

$$K > 0$$

$\therefore$ The value of K lies b/w 0 & 1.553

Ex: For unity feedback system, $G(s) = \frac{K}{s(1+0.4s)(1+0.25s)}$

Find the range of values, marginal value of K & frequency of sustained oscillations.

Sol: charac. eqn is

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(1+0.4s)(1+0.25s)} = 0$$

$$s(1+0.4s)(1+0.25s) + K = 0$$

$$0.1s^3 + 0.65s^2 + s + K = 0$$

$$\begin{array}{c|ccc} s^3 & 0.1 & 1 & \\ \hline s^2 & 0.65 & K & \\ \hline s^1 & \frac{0.65-0.1K}{0.65} & 0 & \\ \hline s^0 & K & 0 & \end{array}$$

For stability of system k should be > 0

$$\therefore k > 0$$

$$\frac{0.65 - 0.115}{0.65} > 0$$

$$\frac{0.65}{0.1} > 15$$

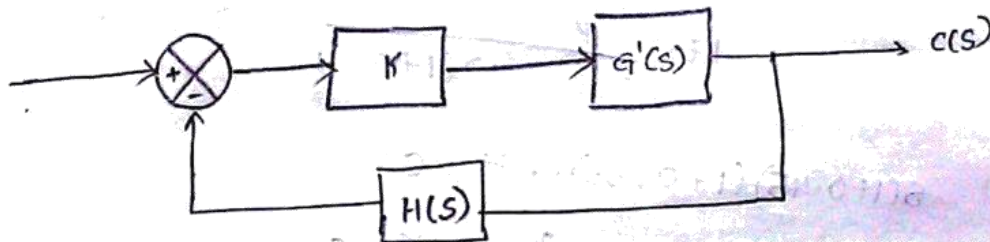
$$0 < k < 6.5$$

* Root Locus :

Movement of closed loop poles with small changes in the parameters of the system can be known by Root Locus method. This method is introduced by W.R. Evans. in 1948.

This is a graphical method in which movement of poles in the s -plane is sketched when a parameter of system is varied from 0 to ∞

* Concept of Root Locus :



Characteristic Eqn

$$1 + G(s)H(s) = 0.$$

$$\text{Where } G(s) = KG'(s)$$

$$\Rightarrow 1 + KG'(s)H(s) = 0.$$

The roots of the above equation are now dependent upon k . If k is varied from $-\infty$ to $+\infty$